

MATHEMATICS XII

10 Year Questions Distribution

According to the Chapters.

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QUESTIONS	Ex. No.	YEARS
Chapter 1: FUNCTION AND LIMITS		
Section B		
4. i) Evaluate any two of the following: a) $\lim_{x \rightarrow 0} \frac{e^{mx} - e^{nx}}{x}$	1.6	2021
b) $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 1}}{x + 1}$ c) $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta}$	1.7	
4. (ii) Evaluate any Two of the following: (a) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ (b) $\lim_{x \rightarrow \infty} \frac{x^2 - 5x + 2}{5x^2 + 6x - 4}$	1.6	2019
4. (ii) Evaluate any Two of the following: (c) $\lim_{x \rightarrow 0} \frac{3e^x - e^{-x} - 2}{x}$	1.7	
4. (ii) Evaluate any Two of the following: (b) $\lim_{x \rightarrow \infty} \frac{x^2 - 5x + 2}{5x^2 + 6x - 4}$		
4. (iv) Two polynomial functions f & g are defined by $f(x) = x^2 - 3x + 4$ and $g(x) = x + 1, \forall x \in \mathbb{R}$ Find $f \circ g, g \circ f$ and show that $f \circ g \neq g \circ f$	1.1	
4. (iv) OR Find the limit of the sequence $\frac{3}{2}, \frac{2}{3}, \frac{5}{4}, \frac{4}{5}, \dots$	1.3	2018
4. (ii) Evaluate any two of the following: (a) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$	1.6	
4. (ii) Evaluate any two of the following: (b) $\lim_{x \rightarrow 0} \frac{e^{mx} - e^{nx}}{x}$	1.7	
4. (ii) Evaluate any two of the following:	Ex.2 1.7	

(c) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$		
4. (v) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 0, & \text{when } x \in Q \\ 1, & \text{when } x \in \mathbb{R} \end{cases}$ Q , being the set of Rationals Find $f(\sqrt{3})$, $f(\pi)$, $f\left(\frac{1}{5}\right)$, $f(2)$ and the value of f at 1.5	1.1	
4. (ii) Evaluate any Two of the following: (a) $\lim_{x \rightarrow 0} \frac{\sin^2\left(\frac{x}{3}\right)}{x^2}$ (b) $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin^3 x}$	1.6	2017
4. (ii) Evaluate any Two of the following: (c) $\lim_{x \rightarrow 0} \frac{3e^x - e^{-x} - 2}{2}$ OR $\lim_{x \rightarrow 0} \left(1 + \frac{3}{t}\right)^t$	1.7	
4. (iv) Two polynomial functions f and g are defined by $f(x) = x^2 - 3x + 4$ and $g(x) = x + 1, \forall x \in \mathbb{R}$. Show that $f \circ g \neq g \circ f$	1.1	
3. (i) Find the limit of the sequence $\frac{1.2}{3.4}, \frac{3.4}{5.6}, \frac{5.6}{7.8}, \dots$	1.3	2016
3. (OR ii) Evaluate any two of the following: (a) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x}$	1.6	
3. (OR ii) Evaluate any two of the following: (b) $\lim_{x \rightarrow 0} \frac{3e^x - e^{-x} - 2}{2}$ (c) $\lim_{x \rightarrow 1} \left[\frac{1}{1-x} - \frac{3}{1-x^3} \right]$	1.7	
3. (ii) Evaluate any two of the following: (a) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x}$	1.3	2015
3. (ii) Evaluate any two of the following: (b) $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 16} - 4}{x}$	1.6	
3. (ii) Evaluate any two of the following: (c) $\lim_{x \rightarrow \infty} \frac{2t^2 - 2t + 1}{t^2 + 4}$	1.7	
[OR 3.(ii)] Determine the limit of the sequence $\frac{1}{1.2}, \frac{1}{2.3}, \frac{1}{3.4}, \dots$	1.3	
3. (ii) Evaluate any tow of the following (a) $\lim_{x \rightarrow a} \frac{x^m - a^m}{x^n - a^n}$	1.6	2014
3. (ii) Evaluate any tow of the following (b) $\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 16} - 4}{x}$	1.6	
3. (ii) Evaluate any tow of the following	1.7	

$$(c) \lim_{x \rightarrow \infty} \frac{\tan x - \sin x}{x}$$

[OR 3.(ii) | Determine the limit of the sequence

$$\frac{1.2}{3.4} + \frac{3.4}{5.6} + \frac{5.6}{7.8} + \dots$$

1.3

3. (ii) Evaluate any Two of the following:

$$(i) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cot \theta - \cos \theta}{\cos^3 \theta}$$

1.6

Q.1

7

2013

(Same Question with different Functions and Value)

3. (ii) Evaluate any Two of the following:

$$(ii) \lim_{x \rightarrow 0} \frac{e^{mx} - e^{nx}}{x}, m, n \in \mathbb{R}$$

1.7

$$(iii) \lim_{x \rightarrow 0} \frac{x}{\sqrt{1 - \cos x}} \quad [\text{OR}(iii)] \lim_{x \rightarrow 1} \left[\frac{1}{1-x} - \frac{3}{1-x^3} \right]$$

[OR 3. (v)] Find the nth term and the limit of the

sequence: $\frac{1.3}{2.4}, \frac{3.5}{4.6}, \frac{5.7}{6.8}, \dots$ where dot '.' Represents multiplication.

1.3

Q.6

(Same Question with different Values)

3. (ii) Evaluate any Two of the following:

$$(i) \lim_{x \rightarrow \infty} \frac{x^2 - 5x + 2}{5x^2 + 6x - 4} \quad (iii) \lim_{x \rightarrow 0} \frac{\operatorname{cosec} \theta - \cot \theta}{\theta}$$

1.6

2012

[OR 3. (v)] Two polynomial functions f and g are defined

by $f(x) = x^2 - 3x + 4$ and fog, gof and show that $fog \neq gof$.

1.1

3. (ii) Evaluate any Two of the following:

$$(ii) \lim_{x \rightarrow 0} \frac{3e^x - e^{-x} - 2}{x}$$

3. (ii) Evaluate any two of the following:

$$(a) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

1.6

2011

3. (ii) Evaluate any two of the following:

$$(b) \lim_{x \rightarrow 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right) \quad (c) \lim_{x \rightarrow 0} \frac{e^{\lambda x} - e^{\mu x}}{x}$$

1.7

$$3. (ii) \text{ Evaluate } \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin^3 x}$$

1.3

2010

Chapter 2: THE STRAIGHT LINE

Section B

2. i) A is two-third the way from (1,10) and (-8,4) and B is the midpoint of (0,-7) and (6,-11), find the distance

2.2

2021

$|AB|$.

i) OR--Find the ratio in which the point $P(0, -1)$ divides the join of $Q(1, 2)$ and $R(2, 5)$.		
2. i) OR If the line through $(2, 5)$ and $(-3, -2)$ is perpendicular to the line through $(4, -1)$ and $(x, 3)$. Find x	2.4	
ii) Determine the equation of the line which passes through the point $(-2, -4)$ and has the sum of its intercepts equal to 3.	2.5	
2. (i) The centroid of triangle, whose two vertices are $(2, 4)$ and $(3, -4)$ is found to be $(3, 1)$ Find its third vertex.	2.2	2019
2. (i) Find the ratio in which y -axis divides the join of $(-5, 3)$ and $(8, 6)$. Also find the coordination of the point of division.	2.2	2018
2. (i) Prove that the points, whose coordinates are $(5, 1), (1, -1)$ and $(11, 4)$, lie on a straight line. Find the intercepts made by this line on the axes.	2.5	2017
[OR 2. (i)] Find the measure of the angle from a line with slope $-2/3$ to : (i) y -axis (ii) x -axis.	2.4	
2. (i) Find the ratio in which y -axis divides the join of $(-5, 3)$ and $(8, 6)$. Also find the coordinates of the point of division.	2.2	2016
2. (ii) Find the equation of the line which passes through the point $(-2, -4)$ and has sum of intercepts equal to 3.	2.5	
2. (ii) Using slopes, prove that $(12, 8), (-2, 6)$ and $(6, 0)$ are the vertices of a right triangle.	2.4	2015
2. (iii) Show that the points $(5, 1), (1, -1)$ and $(11, 4)$ lie on a straight line. Find its equation.	2.5	
2. (i) If the line through $(2, 5)$ and $(-3, -2)$ is perpendicular to the line through $(4, -1)$ and $(x, -3)$, find x .	2.4	2014
2. (i) If the line through $(2, 5)$ and $(-3, -2)$ is perpendicular to the line through $(4, -1)$ and $(x, 3)$, find x .	2.4	2013
2. (ii) Find the equation of the line which passes through the point $(-3, -4)$ and has the sum of its intercepts equal to 1.	2.5	Q. 20
(Same Question With different Values)		
2. (i) A straight line passes through the points $A(-12, -13)$ and $B(-2, -5)$. Find the point on the line whose ordinate is -1 .	2.2	2012
(Same Question With different Values)	Q.24	
2. (ii) Find the equation of a line which passes through the point $(-1, 2)$ and has sum of its intercepts equal to 2.	2.5	Q. 20
(Same Question With different Values)		

2. (i) A is two-third the way from (1,10) to (-8,4) and B is the mid-point of (0,-7), (6,-11). Find the distance $ AB $.	2.2	2011
2. (ii) Find the equation of the straight line which passes through the point (3,4) and makes intercept on the axes such that the y-intercept is twice its x-intercept. (Same Question With different Values)	2.5 Q. 16	
2. (i) If the line through (2,5) and (-3,-2) is perpendicular to the line through (4,-1) and (x,3), find x.	2.4	2010
2. (ii) Find the equation of the line which passes through the point (-3,-4) and has the sum of its intercepts equal to 1. (Same Question With different Values)	2.5 Q. 20	

Section C

7. (a) Three vertices A, B, and C of a triangle are (2, 1), (5, 2) and (3, 4) respectively. Find the coordinates of the circumcenter and the radius of the circumcircle of triangle ABC	2.1	2019
7. (a) Find the equation of the straight line which passes through the point (-3,2) and is such that the portion of it between axes is divided by the point in the ratio 1:2.	2.5	2018
7. (a) A is two-thirds the way from (1,10) to (-8,4) and B is the mid-point of (0, -7) and (6,-11). Find the distance $ AB $	2.2	2017
4. (a) The vertices A, B, and C of a triangle are (2,1), (5,2), & (3,4) respectively. Find the coordinates of the circumcenter and radius of the circumcircle of the triangle ABC.	2.1	2012

Chapter 3: THE GENERAL EQUATION OF STRAIGHT LINES

Section B

iii) The point (2, -5) is a vertex of a square, one of whose sides lies on the line $x - 2y - 7 = 0$. Calculate the area of the square.	3.2	2021
2. (i) [OR] In what ratio is the line segment joining (1, 3) and (2, 7) divided by $3x + y = 9$? (Topic 3.1 Page:101)	Ex.1 3.1	2019
2. (ii) Find the equation of the line which is perpendicular to $2x + 3y + 4 = 0$ and passed through (2, -1). (Topic 3.2 Page:105)	Ex.2 3.1	
2. (iii) A line, whose y-intercept is 1 less than its x-intercept, forms with the coordinate axis a triangle of area 6 square units. What is its equation?	3.2	

2. (iii) Find the coordinates of the foot of the perpendicular from $(-2, 5)$ to $3x + y + 11 = 0$.	3.1	2018
2. (ii) Find the value of k when the vertices of the triangle are the points $(2, 9)$, $(-2, 1)$ and $(k, 3)$ and its area is 28 square unit.	3.2	2017
2. (iii) The gradient of one of the lines $ax^2 + 2hxy + by^2 = 0$ is five times that of the other. Show that $5h^2 = 9ab$.	3.3	
2. (iii) Find the value of k for which the two lines $(k-1)x + ky - 50$, $kx + (2k-1)y + 7 = 0$ intersect at a point lying on the axis of x .	3.1	2016
2. (i) Find the co-ordinates of the foot of perpendicular from $(-2, 5)$ to $3x + y + 11 = 0$	3.1	2015
2. (iv) The area of a triangle is 8 square unit. Two of its vertices are the points $A(1, -2)$ and $B(2, 3)$ and third vertex C lies on the line $2x + y - 2 = 0$ Find co-ordinates of vertex of C .	3.2	
2. (ii) Find the combined equation of the pair of lines through the origin which are perpendicular to the lines represented by $6x^2 - 13xy + 16y^2 = 0$.	3.3	2014
[OR 2. (ii)] The gradient of one of the lines $ax^2 + 2hxy + by^2 = 0$ is two times that of the other. Show that $8h^2 = 9ab$.	3.3	
2. (iii) Find the distance two parallel lines $5x - 12y + 10 = 0$ and $5x - 12y - 16 = 0$.	3.2	
[OR 2. (x)] Find the equation of the locus of the points which are equidistant from the point $(0, 3)$ and the line $y + 3 = 0$ (Same As Question No. 22)		
2. (x) Find the equation of the line passing through the intersection of the line $2x + 3y + 1 = 0$, $3x - 4y - 5 = 0$ and passing through the point $(2, 1)$.	3.1	
2. (iii) Find the value of k when the vertices of the triangle are the points $(2, 6)$, $(6, 3)$ & $(4, k)$ and its area is 15 sq. units.	3.2	2013
2. (iv) The gradient of one of the lines $ax^2 + 2hxy + by^2 = 0$ is five times that of the other. Show that $h^2 = 9ab$.	3.3	
2. (iii) Find the equation of a line through the intersection of the lines $7x - 13y + 46 = 0$, $19x + 11y - 41 = 0$ and passing through the point $(3, 1)$ by using K-method.	3.1	2012
2. (iv) The point $(-2, 1)$ is a vertex of a rectangle whose two sides lie on the lines $3x - 2y - 5 = 0$, $2x + 3y + 7 = 0$ / Find area of the rectangle.	3.2	

2. (iii) The point (2,3) is the foot of the perpendicular dropped from the origin to a straight line. Write the equation of this line.	3.1	2011
2. (iv) Find the distance between the parallel lines $3x + 4y + 10 = 0$, $6x + 8y - 9 = 0$.	Ex.2 3.2	
2. (iii) Find the value of k when the vertices of the triangle are (2,6), (6,3) and (4,k) and its area is 17 square units.	3.2	2010
2. (iv) The gradient of one of the lines $ax^2 + 2hxy + by^2 = 0$ is five times that of the other, show that $5b^2 = 9ab$.	3.3	

Section C

6. a) A line whose y-intercept is 1 less than its x-intercept forms with the co-ordinate axes of triangle of area 6 square units. What is its equation?	3.2	2021
6. a) OR Find the measures of the angles of the triangle, the equations of whose sides are $x + y - 5 = 0$, $x - y + 1 = 0$ and $y = 1$	3.3	
6. (b) The coordinates of two points A and B are (3,4) and (5, -2) respectively. Find the coordinates of any point P if $ PA = PB $ and the area of triangle PAB is 10 square units	3.2	2019
6. (a) Find the centroid of the triangle the equations of whose sides are $12x^2 - 20xy + 7y^2 = 0$ and $2x - 3y + 4 = 0$	3.3	
6. (a) The coordinates of two points A and B are (3,4) and (5,-2) respectively. Find the coordinates of any point P if $ PA = PB $ and ΔPAB is 10 square units.	3.2	2018
[OR 6.(a)] Find the combined equation of the pair of lines through the origin which are perpendicular to the lines represented by $3x^2 + 7xu + 2y^2 = 0$	3.3	
6.(a) The area of a triangle is 8 square units, two of its vertices are the points A(1,-2) and B (2,3) and the third vertex C lies on the line $2x+y-2=0$. Find the coordinates of the vertex C.	3.2	2017
4. (a) Find the equation of the locus of a moving point such that the slope of line joining the point to A(1,3) is three times the slope of the line joining the point B(3,1). (Similar As Question No 21)	3.2	
4. (b) The point (2,-5) is a vertex of a square, one of whose sides lies on the line $x - 2y - 7 = 0$. Calculate the area of the square	3.2	2016

4. (a) Find the equation of the two straight lines passing through (3,-2) and inclined at 60° to the line $\sqrt{3} + x + y = 1$.	3.1	2015
5. (b) If the line $y - 2 = 0$, intersects the pair of lines $x^2 - 7xy + 2y^2 = 0$ in A and B and 'O' be the origin. Find the area of Triangle OAB. (Question No. 9, Old Book of Mathematics)	3.3	
4. (a) Show that the line $x^2 - 4xy + y^2 = 0$ and $x + y = 3$ form an equilateral triangle. Also find the area of the triangle. (Question No. 8, Old Book of Mathematics)	3.3	2011
4. (a)(i) D,E, F are the mid-points of the sides BC, CA, AB respectively of the triangle ABC show that $\Delta ABC = 4\Delta DEF$	3.2	2014
4. (a)(ii) Find the equation of the locus of a moving point such that slope of the line joining the point to A(1,3) is three times that of the slope of the line joining the point to B(3,1). (Similar As Question No 22)	3.2	2010

Chapter 4: DIFFERENTIABILITY

Section B

4.ii) Find the derivative by the first principle at any point x in the Domain D(f) of the function. $f(x) = \sin x^2$ OR $f(x) = x^{\frac{2}{3}}$	4.1	2021
4.iii) Find $\frac{dy}{dx}$ of any Two of the following: a) $y = \frac{1}{2} \tan^2 x + \ln \cos x$ b) $y = (\ln x)^{\sin x}$ c) $y = \ln \left(\frac{e^x}{1+e^x} \right)$	4.2	
4.iv) Find $\frac{dy}{dx}$ of any Two of the following: a) $x^y \cdot y^x = 5$	4.3	
b) $x = a \cos^2 3\theta, y = b \sin^2 3\theta$	4.4	
c) $2x^3 - 3xy + y^3 = 5$ (Implicit Function) (Before Ex.)	4.3	
4.iv) OR --Let $a, b \in \mathbb{R}$ and $y = f(x) = a \cos x + b \sin x,$	4.4	

$\forall x \in \mathbb{R}$. Show that $\frac{d^2y}{dx^2} + y = 0$		
4. (i) Find the derivative by the first principle, at any point $x=a$ in domain $D(f)$ of the function. $f(x) = \cos^2 x$ OR $f(x) = x^{\frac{2}{3}}$	4.1	2019
4. (i) Find the derivative, by the first principle, at $x=a$, in the domain $d(f)$, of the function, $f(x)=\sec x$ OR $f(x) = 3x^2 + x$.	4.1	2018
4. (i) Find the derivative, by the first principle, at $x = a \in D(f)$ of the function $f(x)=\tan x$ OR $f(x) \sin 2x$	4.1	2017
3. (ii) Find the derivate by the first principles at $x=a$ in the domain $D(f)$ of the function $f(x)=\tan x$.	4.1	2016
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: $y(a) = \frac{1}{2} \tan^2 x + \ln \cos x$	4.2	
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (b) $2x^2 - 3xy + y^2 = 5$ (Implicit Function) (Before Ex.)	4.3	
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (c) $x = \ln t + \cos t$, $y = e^t + \sin t$ (Similar as Q.5)	4.4	
3. (i) Find the derivate by First Principle at $x=a$ in the domain of $D(f)$ of $f(x) \operatorname{cosec} x$ OR $f(x)=3x^2+x$	4.1	2015
3. (i) Find the derivate by first principle at $x=a$ is the domain of $d(f)$ of $f(x) = \cot x$ OR $f(x)=3x^2-x$	4.1	2014
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (a) $e^x \ln y = \sin^{-1} y$	4.3	
(b) $\sqrt{x^2 + y^2} = \ln(x^2 - y^2)$	4.3	
(c) $x^y \cdot y^x = 10$	4.3	
3. (i) Find the derivate by the first principles at $x=a$ in the domain $D(f)$ of $f(x)=\sin 2x$	4.1	2013
3. (iii) Find $\frac{dy}{dx}$ of any two of the following: (a) $y = e^{\sin x + \cos x}$ Similar as Q. 4(vii, viii)	4.2	
3. (iii) Find $\frac{dy}{dx}$ of any two of the following: (b) $y = (\sin^{-1} x)^3$	4.2	
3. (iii) Find $\frac{dy}{dx}$ of any two of the following: (c) $y = \sqrt[5]{x^2 + 2x + 3}$	4.2	

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3. (i) Find the derivative by the first principles at any point x in the domain $D(f)$ of the function $f(x) = \cos x^2$	4.1	2012
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (a) $y = (\ln x)^{\tan^{-1} x}$ Similar as Q. 5(iv)	4.2	
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (b) $x^y, y^x = 1$	4.3	
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (c) $x = \sin t^3 + \cos t^3, y = \sin t + 2 \cos^{-1} t$	4.4	
3. (i) Find the derivative by first principles at any point x in the domain $D(f)$ of the function $f(x) = \cot x$	4.1	2011
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (a) $y = \ln \sqrt{1+x^2} + \cot^{-1} x$ Similar question With Different Function	4.2	
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (b) $y = \ln \left(\frac{e^x}{1-e^x} \right)$	4.2	
3. (iii) Find $\frac{dy}{dx}$ of any Two of the following: (c) $x = a \cos^2 3\theta, y = b \sin^2 3\theta$	4.4	
3. (i) Find the derivative by the 1 st principles at $x=a$ in the domain $D(f)$ of $f(x) = \operatorname{cosec} x$	4.1	2010
3. (iii) Find $\frac{dy}{dx}$ of any two of the following: (a) $y = x^{\sin x + \cos x}$ Similar as Q. 4(vii, viii)	4.2	
3. (iii) Find $\frac{dy}{dx}$ of any two of the following: (b) $e^x \ln y = \sin^{-1} y$	4.3	
3. (iii) Find $\frac{dy}{dx}$ of any two of the following: (c) $x = a(\theta - \sin \theta), y = 1 - \cos \theta$ at $\theta = \frac{\pi}{2}$	4.4	
[OR 3. (iii)] if $y = f(x) = a \cos x + b \sin x, \forall x \in \mathbb{R}$ show that $\frac{d^2 y}{dx^2} + y = 0$	4.4	

(Same Question with different variable)		
7. (b) Evaluate $\frac{dy}{dx}$ of any two of the following: (iii) $x = a \cos^2 3\theta$, $y = b \sin^2 3\theta$	4.4 Q.3	
(Same Question with different values)		
7. (b) Evaluate $\frac{dy}{dx}$ of any two of the following: (ii) $x^y \cdot y^x = 10$	4.3	2018
7. (b) Evaluate $\frac{dy}{dx}$ of any two of the following: (iii) $x = \ln t + \sin^{-1} t$, $y = e^t + \cos t$	4.4	
7. (b) Evaluate $\frac{dy}{dx}$ of any two of the following (i) $\sqrt{x^2 + y^2} = \ln(x^2 - y^2)$	4.3	2017
7. (b) Evaluate $\frac{dy}{dx}$ of any two of the following (ii) $y = \cos^{-1} \left(\frac{2x}{1-x^2} \right)$	4.2 Q.5 (ii)	
(Same Question with different function)		
7. (b) Evaluate $\frac{dy}{dx}$ of any two of the following (iii) $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$	4.4	
5. (a) If $y = ae^x + be^{2x} + ce^{3x}$, show that $\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$	4.4	2016

Chapter 5: APPLICATION OF DIFFERENTIAL CALCULUS

Section B

4. (iii) Find the approximate value of $\sin 46^\circ$ using differentials.	5.2	2019
4. (iii) Calculate $\log_{10}(10.1)$ using differentials, given that $\log_{10} e = 0.4343$	5.2	2018
4. (iii) Calculate the approximate value of $\cos 46^\circ$ OR $\cos 59^\circ$, using differentials.	5.2	2017
3. (iv) Using differentials, find the approximate value of $\cos 44^\circ$.		2016
3. (iii) Calculate the approximate value of $\log_{10}(10.1)$, given that $\log_{10} e = 0.4343$.	5.2	2015
3. (iv) Calculate the approximate value of $\sin 46^\circ$ by using differential.		
3. (v) Using differential, show that $\sqrt{x + \Delta x}$ can be approximate to $\sqrt{x} + \frac{1}{2\sqrt{x}} \Delta x$. Hence, find the value of $\sqrt{3.9}$	5.2	2014

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[OR 3.(v)] Calculate an approximate value of $\tan 44^\circ$ by using differential.				
3. (v) Using diff. show that $\sqrt{x + \Delta x}$ can be approximate to $\sqrt{x} + \frac{1}{2\sqrt{x}} dx$. Hence, find the value of $\sqrt{9.1}$			5.2	2013
3. (v) By using differentiations, calculate an approximate value of $\cos 44^\circ$.			5.2	2012
3. (v) Show that $\sqrt{x + \Delta x}$ can be approximated to $\sqrt{x} + \frac{1}{2\sqrt{x}} dx$. Hence, find the value of $\sqrt{3.9}$			5.2	2011
3. (v) Using differential, find the approximate value of $\cos 44^\circ$.			5.2	2010

Section C

7. a) Find the relative maximum or minimum of $f(x) = \frac{x}{\ln x}$			5.3	2021
7. a) OR-- $f(x) = 2e^x + e^{-x}$				
7. (b) [OR] Find the relative maximum and minimum values of the function $f(x) = e^x \sin x$			5.3	2019
[OR 7.(b)] OR Find a right angled triangle of maximum area with a hypotenuse of length h.			5.3	2018
[OR 7. (b)] Find the maximum and minimum value of the function $f(x) = x^3 - 9x^2 + 15x + 3 \forall x \in R$			5.3	2016
5. (b) Find the maximum and minimum values of the function $f(x) = e^x \sin x$				
6. (b) Find the extreme value of the function f given by $f(x) = x(x-1)(x-2) \forall x \in \mathbb{R}$.			5.3 Ex.2	2015
6. (b) Find the relative maximum and minimum value of the function $f(x) = e^x \sin x$ OR $f(x) = x^3 - 9x^2 + 15x + 3$			5.3	2014
6. (b) Find the relative maximum and relative minimum values of the function $f(x) = \frac{\ln x}{x}$			5.3	2013
6. (b) Find the relative maximum and minimum values of the function $f(x) = 2e^x + e^{-x}$			5.3	2012
6. (b) Find the relative maximum and minimum values of the function $f(x) = e^x \sin x$			5.3	2011
6. (b) Show that the maximum value of $f(x) = \frac{\ln x}{x}$ is $\frac{1}{e}$ (Same Question with different values)			5.3 Q.11	2010

Chapter 6: ANTIDERIVATIVES

Section B

4. (v) Solve the differential equation : $2 + 2y \frac{dy}{dx} = 1 + 3x^2$, $y(2) = 1$	6.11	2019
4. (iv) Find the area above the x-axis under the curve $y = 2e^{3x}$, between ordinates $a=2$ and $b=5$.	6.10	2018
[OR 4. (iv)] OR Solve the differential equation: $\frac{dy}{dx} = \sqrt{x \cdot y}$, $x = 100$, $y = 9$	6.11	
4. (v) Find the area above x-axis, under the curve, n $\frac{x^2}{4} + \frac{y^2}{9} = 1$ between the ordinates $x = -1$ and $x = 1$.	6.10	2017
[OR 4.(v) Solve the differential equation $\frac{dy}{dx} = \frac{(1+x^2)^2}{\sin^2 3y}$	6.11	
3. (v) Evaluate any two of the following: (a) $\int_0^2 \frac{dx}{\sqrt{1+x} - \sqrt{x}}$ (Similar as Q. 4)	6.1	2016
3. (v) Evaluate any two of the following: (b) $\frac{\cos \ln x \, dx}{x(3 - \sin \ln x)^{\frac{1}{2}}}$ (Similar as Q. 3(xvi))	6.3	
3. (v) Evaluate any two of the following: (c) $\int \ln x \, dx$	6.7 Ex.2	
3. (v) Evaluate any two of the following: (a) $\int (ax^2 + bx + c)^{-\frac{2}{3}} \cdot \left[x + \frac{b}{c} \right] dx$	6.1	2015
3. (v) Evaluate any two of the following: (b) $\int \cos 5x \cdot \sin 3x \, dx$ (Similar as Q. 5(xi))	6.3	
3. (v) Evaluate any two of the following: (c) $\int \frac{x^2}{\sqrt{1-x^6}} dx$	6.3	
3. (iv) Evaluate any Two of the following: (a) $\int \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right) dx$ (Similar as Q. 22)	6.1	2014
3. (iv) Evaluate any Two of the following: (b) $\int x \ln x \, dx$	6.7	
3. (iv) Evaluate any Two of the following: (c) $\int \frac{\sin(\ln x) dx}{x(3 - \cos \ln x)^{\frac{1}{2}}}$	6.3	

$$(c) \int \cos 4x \cos 2x \, dx$$

3. (iv) Evaluate any Two of the following:

$$(a) \int \sin 4y \sin 2y \, dx$$

(Same Question with Different function)

6.3

Q.N

o

5(xii

i)

6.5

3. (iv) Evaluate any Two of the following:

$$(b) \int_0^a \frac{dx}{(x^2 + a^2)^{\frac{3}{2}}}$$

3. (iv) Evaluate any Two of the following:

$$(c) \int \frac{dy}{\sqrt{4y - y^2}}$$

(Same Question With Different variable)

3. (iv) Evaluate any Two of the following:

$$(a) \int_0^{\pi} \tan^3 x \sec x \, dx$$

$$(b) \int x^2 \tan^{-1} x \, dx$$

$$(c) \int \frac{dx}{4x - x^2}$$

3. (iv) Evaluate any Two of the following:

$$(a) \int x \ln x \, dx$$

$$(b) \int_1^2 (3x^2 + 2x) \sqrt[3]{x^3 + x^2 + 7} \, dx$$

(Similar As Example 1.)

$$(c) \int \sin 3x \cos 5x \, dx$$

$$\text{OR} \int \frac{2x - 3}{x^2 + 2x + 2}$$

Section C

5. Evaluate any Two of the following:

i) $\int \frac{3x^2 + 1}{(x^3 + x - 6)^2} dx$

6.1

2021

ii) $\int \sin^3 x dx$

6.3

iii) $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$

6.5

iv) $\int \frac{1}{25-16y^2} dy$

6.2

7. b) Evaluate any one of the following:

i) $\int x \sin x dx$

6.7

Ex.3

ii) $\int \frac{x^2 dx}{2+x^3-x^6}$

6.9

Ex.6

ii)OR $\int \frac{2x dx}{(1+x^2)(3+x^2)}$

6.9

5. Evaluate any four of the following:

(i) $\int (x^3 + 1)^{\frac{7}{5}} x^5 dx$

6.3

2019

5. Evaluate any four of the following:

(ii) $\int e^x \frac{1 + \sin x}{1 + \cos x} dx$

6.7

5. Evaluate any four of the following:

(iii) $\int_{-6}^{-3} \frac{\sqrt{x^2 - 9}}{x} dx$

6.5

Ex.3

5. Evaluate any four of the following:

(iv) $\int \frac{dx}{x^2 - x + 1}$

6.6

5. Evaluate any four of the following:

(v) $\int \frac{7x - 25}{(x-3)(x-4)} dx$

6.8

(vi) $\int \frac{\sec x \tan x}{a + b \sec x} dx$

6.3

5. Evaluate any four of the following:

(i) $\int_0^2 (x^2 + bx + c)^{-2/3} \left(x + \frac{b}{2}\right) dx$

6.1

2018

5. Evaluate an four of the following:		
(ii) $\int \tan^{-1} x \, dx$	6.7	
5. Evaluate an four of the following:	6.6	
(iii) $\int \frac{dx}{\sqrt{5+4x-x^2}}$		
5. Evaluate an four of the following:	6.8	
(iv) $\int \frac{7x-25}{(x-3)(x-4)} dx$		
5. Evaluate an four of the following:	6.5	
(v) $\int_0^2 \frac{y^3 dy}{\sqrt{16-y^2}}$		
5. Evaluate an four of the following:	6.3	
(vi) $\int \frac{\sec x \tan x}{a+b \sec x} dx$		
7.(b) Evaluate $\frac{dy}{dx}$ of any two of the following:	6.8	
(i) $y = \sqrt{a^2 - x^2} + \ln \sqrt{1+x^2}$		
(Out of the Book)		
5. Evaluate any four of the following:	6.1	2017
(i) $\int_{-1}^1 \frac{3x^2+1}{(x^3+x+6)^{\frac{1}{2}}} dx$		
5. Evaluate any four of the following:	6.6	
(ii) $\int \frac{x+3}{x^2+2x+5} dx$		
5. Evaluate any four of the following:	6.7	
(iii) $\int \sqrt{4-x^2} dx$		
5. Evaluate any four of the following:	6.9	
(iv) $\int \frac{2x-1}{x(x-1)(x-3)} dx$		
5. Evaluate any four of the following:	6.7	
(Using trigonometric substitution) (v) $\int x^2 \tan^{-1} x \, dx$		
5. Evaluate any four of the following:	6.4	
[OR (i)] $\int_{-2}^{-1} x \sqrt{2x^2+3} \, dx$		
(Using trigonometric substitution) (ii) $\int \frac{\pi}{2}$		

[OR 6. (a)] (i) Find the area above the x-axis under the curve $y = \tan x$ between the ordinates $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$	6.10	2016
[OR 6. (a)] (ii) solve the differential equation $\frac{dy}{dx} = \sqrt{xy}$ where $y(9) = 100$.	6.11	
6. (a) Evaluate any two of the following: (i) $\int \frac{\tan x}{\ln(\cos x)} dx$	Misc .Pro bl	
6. (a) Evaluate any two of the following: (ii) $\int \frac{2x}{(1-x^2)(3+x^2)} dx$	6.9	
6. (a) Evaluate any two of the following: (iii) $\int_1^2 (x+1) \sqrt[3]{x^2+2x+2} dx$	Ex. 1 6.4	2015
6. (a) (i) Find the area under the curve $y = 3 \sin x$ between the ordinates $x = 0$ and $x = \frac{\pi}{3}$	6.10	
6. (a) (ii) Solve the differential equation : $\frac{dy}{dx} = x \cos^2 y$.	6.11	
5. (a) Evaluate any Two of the following: (i) $\int \frac{7x-25}{(x-3)(x-4)} dx$	6.8	
5. (a) Evaluate any Two of the following: (ii) $\int x \cdot \tan^{-1} x dx$	6.7	
5. (a) Evaluate any Two of the following: (iii) $\int \frac{x^3}{\sqrt{a^2-x^2}} dx$	6.5	
5. (a) Evaluate any Two of the following: (iv) $\int_0^2 \frac{dx}{9-x^2}$	6.2	
6. (a) (i) Find the area under the curve $y = 3 \sin x$ between the ordinates $x = 0$ and $x = \frac{\pi}{3}$	6.10	
6. (a)(i) Solve the differential equation: $2 + 2y \frac{dy}{dx} = 1 + 3x^2, y(2) = 1$	6.11	2014
[OR 6.(a) (i)] Solve the differential equation: $\frac{du}{dv} = \sqrt{u \cdot v} \quad u = 100, v = 9$	6.11	
6. (a) (ii) Find the area under the curve $y = x - x^2$ between the ordinates $x = 0$ and $x = 1$	6.10	

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5. (a) Evaluate any Two of the following:		6.5		
(i) $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$		6.7		
5. (a) Evaluate any Two of the following:		6.9		
(ii) $\int \frac{2x dx}{\cos^2 2x}$		6.10		
5. (a) Evaluate any Two of the following:		6.11		
(iii) $\int \frac{\sin x dx}{(1+\cos x)(2+\cos x)}$		6.9		
6. (a) (i) Find the area under the curve $y = 3x^4 - 2x^3 + 1$, above x-axis and between $x=1$ and $x=2$.		Ex.3	2013	
6. (a) (ii) Solve the differential equation		6.11		
$\frac{dy}{dx} = y^2 \sin x$		6.9		
5. (a) Evaluate any Two of the following:		6.3		
(a) $\int \frac{\cos x dx}{\sin x (2 + \sin x)}$		6.4		
5. (a) Evaluate any Two of the following:		6.6		
(b) $\int \frac{\sec x \tan x dx}{a + b \sec x}$		6.7		
5. (a) Evaluate any Two of the following:		6.9		
(c) $\int_0^{\frac{\pi}{2}} \cos^4 x dx$		6.4		
5. (a) Evaluate any Two of the following:		6.6	2012	
(i) $\int \frac{dx}{x^2 + 4x + 5}$		6.7		
5. (a) Evaluate any Two of the following:		6.9		
(ii) $\int e^x \frac{1 + \sin x}{1 + \cos x} dx$		6.4		
5. (a) Evaluate any Two of the following:		6.11		
(iii) $\int \frac{\cos x dx}{\sin x (2 + \sin x)}$		Ex.2		
(iii) OR $\int_0^{\pi} \tan^3 x \sec x dx$		6.10		
6. (a)(i) Solve the following differential equation:				
$2 + 2y \frac{dy}{dx} = 1 + 3x^3, y(2) = 1$				
6. (a) (ii) Find the area above x-axis under the circle				
$x^2 + y^2 = 4$ and between the ordinates $x = 1$ and $x = 3$				

6. (a) (ii) Find the area above x-axis under the curve $y = 3x^4 - 2x^2 + 1$ & between the ordinates $x=1$ and $x=2$.	6.10	2011
5. (a) Evaluate any Two of the following: (i) $\int \frac{2x dx}{(1+x^2)(3+x^2)}$	6.9	
5. (a) Evaluate any Two of the following: (ii) $\int \frac{x^3 dx}{\sqrt{a^2 - x^2}}$	6.5 Q. 6	
Same Question With different Value		
5. (a) Evaluate any Two of the following: (iii) $\int \frac{\sec x \tan x dx}{a + b \sec x}$	6.3	
6. (a) (i) Solve the following differential equation: $\frac{dy}{dx} = \frac{\sqrt{1 + \cos y}}{\sin y}, y(3) = \frac{\pi}{2}$	6.1	2011
5. (a) Evaluate any Two of the following (i) $\int x^2 \sqrt{4+x} dx$	6.3	2010
5. (a) Evaluate any Two of the following (ii) $\int \frac{\cos x dx}{\sin x (2 + \sin x)}$	6.9	
5. (a) Evaluate any Two of the following (iii) $\int \frac{\tan x}{\ln \cos x} dx$	Misc Prob I	
6. (a) (i) Solve the differential equation $y \frac{dy}{dx} = x(y^4 + 2y^2 + 1), y(-3) = 1$	6.11	
6. (a) (ii) Find the area above the x-axis between the ordinates $x = \frac{\pi}{4}$ & $x = \frac{\pi}{3}$ under the curve $y = \tan x$	6.10	

Chapter 7:

CIRCLE

Section B

3. i) Find the equation of the circle having $(-5, 6)$ and $(3, -4)$ are the end points of a diameter	7.1	2021
3. i) OR Find the equation of the circle passing through the points $(0, 3)$, $(2, -1)$ and $(1, 0)$.		
3. ii) Find the equation of the circle with radius $\sqrt{a^2 + b^2}$ which passes through the two points $(a, 0)$ and $(-a, 0)$.		
3. (i) Find the equation of the circle of radius a which passes through the two points on the x-axis which are at a distance b from the origin.	7.1	2019

3. (i) Find the equation of the circle concentric with the circle $x^2 + y^2 + 6x - 10y + 33 = 0$ and touching the line $y=0$	7.1	2018
3. (ii) Prove that the product of abscissas of the point where the straight line $y=mx$ meets the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is equal to $\frac{c}{1+m^2}$.	7.2	
3. (i) Find the equation of the circle containing the points $(-1,-1)$ and $(3,1)$ and with the Centre on the line $x - y + 10 = 0$	7.1	2017
3. (ii) Show that lines $x=5$ and $y=7$ both touch the circle $x^2 + y^2 - 4x - 8y + 11 = 0$.	7.2	
2. (iv) Prove that the point $(5,-7,5)$ lies outside the circle whose equation is $x^2 + y^2 - 4x + 2 = 44$	7.1	2016
2. (v) Find the equation of a circle with centre at the point $(1,-1)$ and touching the straight line $5x - 2y = 7$.	7.2	
2. (v) Find the equation of the circle containing the points $(-1,-1)$ and $(3,1)$ and the line $x - y + 10 = 0$ passing through the centre of the circle	7.1	2015
2. (vii) Prove that the two circle $x^2 + y^2 + 2gx + 2fy + c = 0$ and $x^2 + y^2 + 2fx + 2gy + c = 0$ touch each other if $\frac{1}{f^2} + \frac{1}{g^2} = \frac{1}{c}$	7.2	
2. (v) Find the equation of the circle which is concentric with the circle $x^2 + y^2 - 8x + 12y - 12 = 0$ and passing through the point $(5,4)$.	7.2	2014
2. (v) Find the equation of the circle touching each of the axes in 4 th quadrant at a distance of 5 units from the origin.	7.1	2013
2. (vi) Find the equation of the circle which is concentric with the circle $x^2 + y^2 + 8x + 12y + 15 = 0$ and passes through the point $(5,4)$	7.2	
2. (v) Find the equation of circle concentric with the circle $x^2 + y^2 + 6x - 10y + 33 = 0$ and touching the y-axis.	7.1	2012
2. (vi) Prove that the straight line $y = x + c\sqrt{2}$ touches the circle $x^2 + y^2 = c^2$ and find its point of contact.	7.2	
2. (v) Find the equation of circle which passes through the origin and cuts off intercepts equal to 3 and 4 from the axes.	7.1	2011
2. (vi) Prove that the curves $x^2 + 3y^2 - 24 = 0$ and $3x^2 - y^2 = 12$ intersect at right angle at the point $(\sqrt{6}, \sqrt{6})$.	7.2	

2. (vii) Find the equation of the circle touching each of the axes in 4 th quadrant at a distance of 6 units from the origin.	7.1	2010
2. (viii) Find the equation of the circle which is concentric with the circle $x^2 + y^2 - 8x + 12y - 12 = 0$ and passes through the point (5,4).	7.2	

Section C

4. (b) Find the condition that conics $ax^2 + by^2 = 1$ and $a'x^2 + b'y^2 = 1$ cut each other orthogonally.	7.2	2014
4. (a) Equation of a curve is given by $x^2 - 2xy + y^2 + 2x - 4 = 0$ find the slope of the curve at the point (2,2). (Out of the Book)	7.2	2013
4. (b) find Equation of the circle containing the points (-1, -1) and (3, 1) and with the centre on the line $x - y + 10 = 0$.	7.1	
4. (b) Find the condition that conic $ax^2 + by^2 = 1$ should cut $a'x^2 + b'y^2 = 1$ orthogonally.	7.2	2012
4. (b) Find the equation of circle containing the point (-1, -1) and (3, 1) and with centre on the line $x - y + 10 = 0$.	7.1	2011
4. (b) Prove that two circle $x^2 + y^2 + 2gx + 2fy + c = 0$ and $x^2 + y^2 + 2f'y + c' = 0$ touch each other, if $\frac{1}{f^2} + \frac{1}{g^2} = \frac{1}{c}$	7.2	2010

Chapter 8: PARABOLA, ELLIPSE AND HYPERBOLA

Section B

3. iii) Find the equation of tangent and normal to the hyperbola $x^2 - y^2 = 49$ at (8, 15).	8.4	2021
3. iii) (OR) Find p, if the line $2\sqrt{2} - 3y = p$ touch the hyperbola $16x^2 - 36y^2 = 576$		
3. iv) Show that the eccentricities e_1 and e_2 of the tow conjugate hyperbolas satisfy the relation $e_1^2 + e_2^2 = e_1^2 e_2^2$		2019
3. (ii) Find the equation of the parabola with focus (3,4) and directrix $x + y - 1 = 0$.	8.1	

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3. (iv) Find the eccentricity, foci & directrices of Hyperbola $9x^2 - y^2 + 1 = 0$.	8.3	2018	
3. (v)[OR] Show that the eccentricities e_1 & e_2 of two conjugate hyperbolae satisfy the relation: $e_1^2 + e_2^2 = e_1^2 e_2^2$.	8.4		
3. (iii) Prove that the focal radius of the point (a,b) on the parabola $x^2 = 4ay$ equals $ a + b $	Ex. 4 8.1		
[OR 3. (iii)] Find the equation of the parabola whose vertex is (3,2) and the ends of focal chord are (5,6) and (5,-2)	8.1		
3. (iv) Find the eccentricity of the ellipse whose axes are 32 and 24.	8.2		
3. (v) Find the equation of the rectangular hyperbola with Centre at (0,0) and vertices (0,±3).	8.3	2017	
3. (iii) Find the equation of the parabola whose focus is (3,4) and the directrix is the line $x + y - 1 = 0$.	8.1		
3. (iv) Find the Centre, foci and equations of directrix of the hyperbola $\frac{(x+2)^2}{4} - \frac{(y-1)^2}{9} = 1$	8.3		
3. (v) Find the eccentricity of an ellipse whose latus rectum is equal to half of its major axes. OR 3(v) The length of the major axis of an ellipse is 25 & its foci are points (±5, 0). Find the equation of the ellipse.	8.2		
2. (vi) Find the equation of the parabola with focus (2,3) and directrix $y - 5 = 0$	8.1	2016	
2. (vii) The length of the major axis of an ellipse is 25 units and its foci are the points (± 5, 0) . Find its equation.	8.2		
2. (viii) Find the eccentricity, foci , directrices and length of the latus rectum of the hyperbola $9x^2 - y^2 + 1 = 0$.	8.3		
2. (vi) Find the eccentricity of an ellipse whose length of latus rectum is half of its major axes.	8.2	2015	
2. (viii) Find centre, foci and equation of directrices of hyperbola $\frac{(x-1)^2}{4} - \frac{(y-2)^2}{9} = 1$	8.3		
[OR (viii)] Find the equation of circle whose diameter is the length of latus rectum of parabola $x^2 = 36y$.	8.1		
2. (iv) Find the equation of the parabola having focus (-5,3) and directrix $y - 7 = 0$	8.1	2014	

2. (vi) Find the centre, focus and eccentricity of the ellipse $\frac{(x-3)^2}{25} + \frac{(y+1)^2}{9} = 1$	8.2	
2. (vii) Find the equation of hyperbola with focus (8,0) and directrix $x=4$	8.3	
2. (vii) Determine the vertex, focus and equation of directrix of the curve $x^2 + 4x + 4y - 12 = 0$	8.1	2013
2. (viii) Find equation of the hyperbola having focus (8,0) and directrix $x=4$.	8.3	
[OR 2. (viii)] Find the eccentricity, foci and equations of directrices of $25x^2 + 9y^2 = 225$	8.2	
2. (vii) Find the equation of parabola with focus (2,3) and directrix $y - 5 = 0$	8.1	2012
2. (viii) Find the equation of ellipse whose centre is at (0,0), $e = \frac{2}{3}$, latus rectum of length $\frac{20}{3}$ and major axis is along x-axis	8.2	
[OR 2. (viii)] Find the eccentricity, foci and equation of directrices of hyperbola $9x^2 - y^2 + 1 = 0$.	8.3	
2. (vii) Determine the vertex, focus and equation of directrix of the parabola $x^2 + 4x + 4y - 12 = 0$.	8.1	2011
2. (viii) Find the eccentricity, foci and equation of directrices of the hyperbola $16x^2 - 9y^2 = 144$.	8.3	
[OR 2. (viii)] The length of the major axis of an ellipse is 20 and its foci are the points $(\pm 5, 0)$. Find the equation of the ellipse.	8.2	
2. (v) Find the equation of the circle whose diameter is the latus rectum of the parabola $y^2 = -36x$.	8.2	2010
2. (vi) Find the eccentricity, foci and equation of directrices of $25x^2 + 9y^2 = 225$.	8.3	
[OR 2. (vi)] Find the eccentricity of the hyperbola whose latus rectum is for times that of the transverse axis.	8.3	
Section C		
6. b) Prove that the line $lx + my + n = 0$ and the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ have just one point in common if $a^2l^2 + b^2m^2 - n^2 = 0$	8.4	2021

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3. (v)[OR] Show that the eccentricities e_1 & e_2 of two conjugate hyperbolae satisfy the relation: $e_1^2 + e_2^2 = e_1^2 e_2^2$.	8.4	2019		
6. (b) Find the equation of the tangents at the ends of the latus rectum of the parabola $y^2 = 4ax$	8.4	2018		
6. (b) Prove that the line $lx + my + n = 0$ and the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ have just one point in common, if $a^2 l^2 + b^2 m^2 - n^2 = 0$.	8.4	2017		
6. (b) Prove that the parabolas $x^2 = 4ay$ and $y^2 = 4ax$ intersect at angle $\theta = \tan^{-1} \frac{3}{2} \left[\frac{\frac{1}{a^3} \frac{1}{b^3}}{\frac{2}{a^3} \frac{2}{b^3}} \right]$	8.4	2016		
4. (b) Find the equation of circle passing through the focus of parabola $x^2 + 8y = 0$ and foci of ellipse, $16x^2 + 25y^2 = 400$.	8.2	2015		
[OR 4. (b)] Prove that the line $lx + my + n = 0$ and the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ have just one point in common if $a^2 l^2 + b^2 m^2 - n^2 = 0$	8.4			
4. (a) Find the equation of circle passing through the focus of parabola $x^2 + 8y = 0$ and foci of ellipse $16x^2 + 25y^2 = 400$.	8.2	2014		
[OR 4. (a)] Determine the vertex, focus and the equation of directrix of $y^2 + 4y + 3x - 92 = 0$	8.1			
5. (b) Show that the eccentricities e_1 and e_2 of two conjugate hyperbola satisfy the relation $e_1^2 + e_2^2 = e_1^2 e_2^2$	8.4			
[OR 5. (b)] if $y = \sqrt{5}x + k$ is a tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$. What is k?				
5. (b) Prove that line $lx + my + n = 0$ & the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ have just one point in common if $a^2 l^2 + b^2 m^2 - n^2 = 0$	8.4	2013		
5. (b) Prove that the angle between the conics $x^2 = 4ay$ and $y^2 = 4bx$ at a point other than the origin $\theta = \tan^{-1} \frac{3}{2} \left[\frac{\frac{1}{a^3} \frac{1}{b^3}}{\frac{2}{a^3} \frac{2}{b^3}} \right]$	8.4	2012		
5. (b) Find the eccentricity, centre, vertices and the foci of the ellipse given by the equation/ $4x^2 - 16x + 25y^2 + 200y + 316 = 0$.	8.2	2011		
5. (b) Show that the eccentricities e_1 and e_2 of two conjugate hyperbola satisfy the relation $e_1^2 + e_2^2 = e_1^2 e_2^2$	8.4	2010		

Chapter 9:

VECTORS

Section B

iv) A particle acted upon by the force $4\hat{i} + 2\hat{j} - 3\hat{k}$ and $3\hat{i} + \hat{j} - \hat{k}$ is displaced from the point P(1, 2, 3) to the point Q(5, 4, 1); find the work done on the particle.	9.7	2021
iv) OR--Find $\sin(a, b)$ for the vectors $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.	9.5	
2. (iv) A particle, acted upon by the forces $4\hat{i} + \hat{j} - 3\hat{k}$ and $3\hat{i} + \hat{j} - \hat{k}$, is displaced from point (1,2,3) to point (4,4,1). Find the work done	9.7	2019
2. (iv) [OR] Calculate $\sin(\vec{a}, \vec{b})$ for vectors $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$	9.5	
2. (v) Prove that $ \vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a} = 2 \vec{a} \quad \vec{b} \quad \vec{a} $	9.6	
2. (iv) Calculate $\sin(\vec{a}, \vec{b})$ for vector $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$. OR Find $\cos \theta$, where θ is the angle between the vectors $2\hat{i} + \hat{j} - 3\hat{k}$ and $-\hat{i} + 4\hat{j} - 2\hat{k}$	9.5	2018
2. (v) simplify: $[\vec{a}, 2\vec{b} - 3\vec{c}, 2\vec{a} + \vec{b} + \vec{c}]$	9.6	
2. (iv) Find the unit vector perpendicular to both of the vectors $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} - 4\hat{k}$. Also calculate $\sin$ of the angle between the given vectors. [OR 2. (iv)] Find $\cos \theta$, where θ is the angle between vectors $2\hat{i} + \hat{j} - 3\hat{k}$ and $-\hat{i} + 4\hat{j} - 2\hat{k}$.	9.5	2017
2. (v) Resolve the vector $\vec{a} = (6, 8, -6)$ in the direction of vectors $\vec{p}_1 = (1, -1, 2)$, $\vec{p}_2 = (2, 2, -1)$ and $\vec{p}_3 = (3, 7, -7)$	9.3	
2. (ix) A Particle is acted upon by constant forces $4\hat{i} + \hat{j} - 3\hat{k}$ and $3\hat{i} + \hat{j} - \hat{k}$ and displaced from the point $\hat{i} + 2\hat{j} - 3\hat{k}$ to the point $5\hat{i} + 4\hat{j} - \hat{k}$. Find the work done by the forces.	9.7	2016
2. (x) Prove that $ \vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a} = 2 \vec{a} \quad \vec{b} \quad \vec{c} $	9.6	
2. (x OR) Find the angle between the positive y-axis and the vector $\vec{a} = -\hat{i} + 3\hat{j} + 2\hat{k}$	9.3	
2. (ix) Find constant "a" such that the set of vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$, $3\hat{i} + a\hat{j} + 5\hat{k}$ are coplanar.	9.6	2015